1.22 Exponentials and Logarithms

Worked Solutions

1 a)

$4^x = 3 \Rightarrow x = \log_4 3.$

1 b)

$3^x = 4 \Rightarrow x = \log_3 4.$

1 c)

$2^x = \frac{1}{3} \Rightarrow x = \log_2 \frac{1}{3} = -\log_2 3.$

1 d)

$9^x = \frac{1}{100} \Rightarrow x = \log_9 \frac{1}{100} = -\log_9 100.$

2 a)

$2^{x+1} = 3 \Rightarrow x + 1 = \log_2 3 \Rightarrow x = \log_2 3 - 1.$

2 b)

$3^{x-5} = \frac{1}{5} \Rightarrow x - 5 = \log_3 \frac{1}{5} \Rightarrow x = \log_3 \frac{1}{5} + 5.$

2 c)

$5^{2x-1} = 7^{500} \Rightarrow 2x - 1 = \log_5 7^{500}$
$\Rightarrow 2x = 500 \log_5 7 + 1 \Rightarrow x = 250 \log_5 7 + \frac{1}{2}.$

2 d)

$6^{3x+4} = 9^{250} \Rightarrow 3x + 4 = \log_6 9^{250}$
$\Rightarrow 3x = 250 \log_6 9 - 4 \Rightarrow x = \frac{250}{3} \log_6 9 - \frac{4}{3}.$

3)

$6^{x^2-5} = \frac{1}{216} \Rightarrow x^2 - 5 = \log_6 \frac{1}{216}$
$\Rightarrow x^2 = \log_6 \frac{1}{216} + 5 \Rightarrow x^2 = \log_6 6^{-3} + 5$
$\Rightarrow x^2 = -3 + 5 \Rightarrow x^2 = 2 \Rightarrow x = \pm\sqrt{2}.$

4)

$8^{10-x^2} = 32768 \Rightarrow 10 - x^2 = \log_8 32768$
$\Rightarrow x^2 = 10 - \log_8 32768 \Rightarrow x^2 = 10 - \log_8 8^5$
$\Rightarrow x^2 = 10 - 5 \Rightarrow x^2 = 5 \Rightarrow x = \pm\sqrt{5}.$

5 a)

$\log_3 x = 8 \Rightarrow x = 3^8 = 6561.$

5 b)

$\log_8 x = 3 \Rightarrow x = 8^3 = 512.$

5 c)

$\log_5 x = 7 \Rightarrow x = 5^7 = 78125.$

5 d)

$\log_7 x = 5 \Rightarrow x = 7^5 = 16807.$

6 a)

$\log_3 4 + \log_3 5 = \log_3 20.$

6 b)

$\log_5 3 + \log_5 6 = \log_5 18.$

6 c)

$\log_6 10 - \log_6 2 = \log_6 5.$

6 d)

$\log_{10} 4 - \log_{10} 8 = \log_{10} \frac{1}{2}$
$(= \log_{10} 2^{-1} = -\log_{10} 2).$

7 a)

$\log_{10} 4 + \log_{10} 3 - \log_{10} 2 = \log_{10} 12 - \log_{10} 2$
$= \log_{10} 6.$

7 b)

$\log_2 20 - \log_2 6 + \log_2 3 = \log_2 \frac{10}{3} + \log_2 3 = \log_2 10.$

7 c)

$\log_5 9 - \log_5 10 - \log_5 3 = \log_5 \frac{9}{10} - \log_5 3 = \log_5 \frac{3}{10}.$

7 d)

$\log_9 8 - \log_9 10 + \log_9 4 = \log_9 \frac{4}{5} + \log_9 4 = \log_9 \frac{16}{5}.$

8)

$\log_3 x - \log_3 5 = \log_3 8 \Rightarrow \log_3 \frac{x}{5} = \log_3 8 \Rightarrow \frac{x}{5} = 8$
$\Rightarrow x = 40.$

9)

$\log_7 9 - \log_7 x = \log_7 18 \Rightarrow \log_7 \frac{9}{x} = \log_7 18$
$\Rightarrow \frac{9}{x} = 18 \Rightarrow x = \frac{1}{2}.$

10 a)

$\log_3 3 = 1.$

10 b)

$\log_8 8 = 1.$

10 c)

$2 \log_5 5 = 2 \times 1 = 2.$

10 d)

$-4 \log_{10} 10 = -4 \times 1 = -4.$

11 a)

$2 \log_3 5 = \log_3 5^2 = \log_3 25.$

11 b)

$3 \log_4 3 = \log_4 3^3 = \log_4 27.$

11 c)

$4 \log_9 2 = \log_9 2^4 = \log_9 16.$

11 d)

$5 \log_7 10 = \log_7 10^5 = \log_7 100000.$

12 a)

$-\log_2 3 = \log_2 3^{-1} = \log_2 \frac{1}{3}.$

12 b)

$-\log_3 2 = \log_3 2^{-1} = \log_3 \frac{1}{2}.$

12 c)

$-2 \log_5 2 = \log_5 2^{-2} = \log_5 \frac{1}{4}.$

12 d)

$-3 \log_8 3 = \log_8 3^{-3} = \log_8 \frac{1}{27}.$

13 a)

$\log_b 2 + 2 \log_b 5 = \log_b 2 + \log_b 25 = \log_b 50.$

13 b)

$\log_b 5 + 3 \log_b 3 = \log_b 5 + \log_b 27 = \log_b 135.$

13 c)

$\log_b 10 - 4 \log_b 2 = \log_b 10 - \log_b 16 = \log_b \frac{5}{8}.$

13 d)

$\log_b 4 - 2\log_b 10 = \log_b 4 - \log_b 100 = \log_b \frac{1}{25}$.

14 a)

$3\log_2 x + \log_2 k = \log_2 x^3 + \log_2 k = \log_2 kx^3$.

14 b)

$\log_2 kx^3 = 3 \Rightarrow kx^3 = 8 \Rightarrow x^3 = \frac{8}{k} \Rightarrow x = \sqrt[3]{\frac{8}{k}} = \frac{2}{\sqrt[3]{k}}$.

15 a)

$5\log_3 x - \log_3 m = \log_3 x^5 - \log_3 m = \log_3 \frac{x^5}{m}$.

15 b)

$\log_3 \frac{x^5}{m} = 2 \Rightarrow \frac{x^5}{m} = 9 \Rightarrow x^5 = 9m \Rightarrow x = \sqrt[5]{9m}$.

16 a)

$\log_5 30 - 2\log_5 2 = \log_5 30 - \log_5 4 = \log_5 \frac{15}{2}$.

16 b)

$\log_2 9 - 3\log_2 3 = \log_2 9 - \log_2 27 = \log_2 \frac{1}{3}$.

16 c)

$\log_9 100 + 2\log_9 5 = \log_9 100 + \log_9 25 = \log_9 2500$.

16 d)

$\log_3 20 + 3\log_3 10 = \log_3 20 + \log_3 1000$
$= \log_3 20000$.

17)

$\log_5 y^5 - \log_5 x^3 = 10 \Rightarrow \log_5 \frac{y^5}{x^3} = 10 \Rightarrow \frac{y^5}{x^3} = 5^{10}$
$\Rightarrow y^5 = 5^{10}x^3 \Rightarrow y = 25x^{\frac{3}{5}}$.

18 a)

$2\log_3 4 + 4\log_3 2 = \log_3 16 + \log_3 16 = \log_3 256$.

18 b)

$3\log_5 4 - 2\log_5 2 = \log_5 64 - \log_5 4 = \log_5 16$.

18 c)

$3\log_8 5 + 2\log_8 10 = \log_8 125 + \log_8 100$
$= \log_8 12500$.

18 d)

$4\log_4 3 - 3\log_4 9 = \log_4 81 - \log_4 729 = \log_4 \frac{1}{9}$.

19 a)

$\frac{1}{2}\log_7 4 + \frac{1}{3}\log_7 8 = \log_7 2 + \log_7 2 = \log_7 4$.

19 b)

$\frac{1}{2}\log_{10} 81 - \frac{1}{3}\log_{10} 125 = \log_{10} 9 - \log_{10} 5 = \log_{10} \frac{9}{5}$.

19 c)

$\frac{2}{3}\log_2 27 + \frac{3}{2}\log_2 9 = \log_2 9 + \log_2 27 = \log_2 243$.

19 d)

$\frac{4}{5}\log_9 32 - \frac{5}{4}\log_9 16 = \log_9 16 - \log_9 32 = \log_9 \frac{1}{2}$.

20 a)

$4\log_3 2 + 1 = \log_3 16 + \log_3 3 = \log_3 48$.

20 b)

$3\log_2 5 - 1 = \log_2 125 - \log_2 2 = \log_2 \frac{125}{2}$.

20 c)

$2\log_5 6 + 2 = \log_5 36 + \log_5 25 = \log_5 900$.

20 d)

$3 - 2\log_4 10 = \log_4 64 - \log_4 100 = \log_4 \frac{16}{25}$.

21 a)

$2\log_b 3 + 2 = \log_b 9 + \log_b b^2 = \log_b 9b^2$.

21 b)

$4\log_b 2 - 1 = \log_b 16 - \log_b b = \log_b \frac{16}{b}$.

21 c)

$3\log_b 4 + 1 = \log_b 64 + \log_b b = \log_b 64b$.

21 d)

$1 - 2\log_b 20 = \log_b b - \log_b 400 = \log_b \frac{b}{400}$.

22)

$\log_b p - 2\log_b q = \log_b(p - 2q)$
$\Rightarrow \log_b p - \log_b q^2 = \log_b(p - 2q)$
$\Rightarrow \log_b \frac{p}{q^2} = \log_b(p - 2q) \Rightarrow \frac{p}{q^2} = p - 2q$
$\Rightarrow p = pq^2 - 2q^3 \Rightarrow p - pq^2 = -2q^3$
$\Rightarrow p(1 - q^2) = -2q^3 \Rightarrow p = -\frac{2q^3}{1-q^2} \Rightarrow p = \frac{2q^3}{q^2-1}$.

23)

$3\log_b q - \frac{1}{2}\log_b p = \log_b(3q-p)$
$\Rightarrow \log_b q^3 - \log_b p^{\frac{1}{2}} = \log_b(3q-p)$
$\Rightarrow \log_b \frac{q^3}{p^{\frac{1}{2}}} = \log_b(3q-p) \Rightarrow \frac{q^3}{p^{\frac{1}{2}}} = 3q-p$
$\Rightarrow p^{\frac{1}{2}} = \frac{q^3}{3q-p} \Rightarrow p = \frac{q^6}{9q^2-6pq+p^2}$.

24 a)

$\log_b b^2 + \log_b b^5 = 2+5 = 7$.

24 b)

$\log_b b^4 - \log_b \frac{1}{b} = \log_b b^4 - \log_b b^{-1} = 4-(-1)$
$= 5$.

24 c)

$\log_b \frac{1}{b^2} + \log_b b^3 = \log_b b^{-2} + \log_b b^3 = -2+3$
$= 1$.

24 d)

$\log_b \frac{1}{b} - \log_b \frac{1}{b^3} = \log_b b^{-1} - \log_b b^{-3}$
$= (-1)-(-3) = 2$.

25)

$\log_3 \frac{x^3}{9} + \log_3 27x - \log_3 x^5$
$= \log_3(\frac{x^3}{9} \times 27x) - \log_3 x^5 = \log_3 3x^4 - \log_3 x^5$
$= \log_3(\frac{3x^4}{x^5}) = \log_3 \frac{3}{x}$.

26 a)

$\log_4 \sqrt{x} = \log_4 x^{\frac{1}{2}} = \frac{1}{2}\log_4 x = \frac{1}{2}m$.

26 b)

$\log_4 \frac{x}{4} = \log_4 x - \log_4 4 = m-1$.

26 c)

$\log_4 \frac{16}{x^2} = \log_4 16 - \log_4 x^2 = 2\log_4 4 - 2\log_4 x$
$= 2-2m$.

27)

$\log_b y = 3\log_b 2 + \log_b 5 - \frac{3}{2}$
$\Rightarrow \log_b y = \log_b 8 + \log_b 5 - \log_b b^{\frac{3}{2}}$
$\Rightarrow \log_b y = \log_b \frac{40}{b^{\frac{3}{2}}} \Rightarrow y = \frac{40}{b^{\frac{3}{2}}} = 40b^{-\frac{3}{2}}$.

28)

$\log_k y = \log_k 8 - 2\log_k \frac{1}{2} + \frac{3}{4}$
$\Rightarrow \log_k y = \log_k 8 - \log_k \frac{1}{4} + \log_k k^{\frac{3}{4}}$
$\Rightarrow \log_k y = \log_k 32k^{\frac{3}{4}} \Rightarrow y = 32k^{\frac{3}{4}}$.

29 a)

$2\log_b p + 3\log_b q + 2\log_b r = \log_b p^2 + \log_b q^3 + \log_b r^2 = \log_b(p^2q^3r^2)$.

29 b)

$3\log_b p + 5\log_b q - 3\log_b r = \log_b p^3 + \log_b q^5 - \log_b r^3 = \log_b(\frac{p^3q^5}{r^3})$.

29 c)

$5\log_b p - 2\log_b q + 4\log_b r = \log_b p^5 - \log_b q^2 + \log_b r^4 = \log_b(\frac{p^5r^4}{q^2})$.

29 d)

$8\log_b p - 6\log_b q - 9\log_b r$
$= \log_b p^8 - \log_b q^6 - \log_b r^9 = \log_b(\frac{p^8}{q^6r^9})$.

30 a)

$\log_7 30 - (\log_7 2 + \log_7 3) = \log_7 30 - \log_7 6$
$= \log_7 5$.

30 b)

$\log_8 15 - 2(\log_8 6 - \log_8 2) = \log_8 15 - 2\log_8 3$
$= \log_8 15 - \log_8 9 = \log_8 \frac{5}{3}$.

30 c)

$\log_6 60 - 3(\log_6 5 + \log_6 2) = \log_6 60 - 3\log_6 10$
$= \log_6 60 - \log_6 1000 = \log_6 \frac{3}{50}$.

30 d)

$\log_{10} 50 - 2(\log_{10} 80 - \log_{10} 16)$
$= \log_{10} 50 - 2\log_{10} 5 = \log_{10} 50 - \log_{10} 25$
$= \log_{10} 2$.

31 a)

$\log_b p - (\log_b q + \log_b r) = \log_b p - \log_b qr$
$= \log_b(\frac{p}{qr})$.

31 b)

$\log_b p - 2(\log_b q - \log_b r) = \log_b p - 2\log_b \frac{q}{r}$
$= \log_b p - \log_b \frac{q^2}{r^2} = \log_b(\frac{pr^2}{q^2})$.

31 c)

$\log_b p - 3(\log_b q + \log_b r) = \log_b p - 3\log_b qr$
$= \log_b p - \log_b q^3r^3 = \log_b(\frac{p}{q^3r^3})$.

31 d)

$2\log_b p - 2(\log_b q - 2\log_b r)$
$= \log_b p^2 - 2(\log_b q - \log_b r^2)$

$= \log_b p^2 - 2\log_b \frac{q}{r^2} = \log_b p^2 - \log_b \frac{q^2}{r^4}$
$= \log_b(\frac{p^2r^4}{q^2})$.

32 a)

$\log_b 15 = \log_b 3 + \log_b 5 = x + y$.

32 b)

$\log_b 9 = 2\log_b 3 = 2x$.

32 c)

$\log_b 21 = \log_b 3 + \log_b 7 = x + z$.

32 d)

$\log_b \frac{49}{25} = 2\log_b \frac{7}{5} = 2(\log_b 7 - \log_b 5) = 2(z - y)$
$= 2z - 2y$.

33)

$\log_3 4 \times \log_3 64 = \log_3 4 \times 3\log_3 4 = 3(\log_3 4)^2 > 3$ as $\log_3 4 > 1$.

34 a)

$\log_{100} 10 + \log_{10} 100 = \log_{100} 100^{\frac{1}{2}} + \log_{10} 10^2$
$= \frac{1}{2} + 2 = \frac{5}{2}$.

34 b)

$\log_2 8 - \log_8 2 = \log_2 2^3 - \log_8 8^{\frac{1}{3}} = 3 - \frac{1}{3} = \frac{8}{3}$.

34 c)

$\log_9 81 + \log_3 \frac{1}{3} = \log_9 9^2 + \log_3 3^{-1} = 2 - 1 = 1$.

34 d)

$\log_{625} \frac{1}{625} + \log_5 625 = \log_{625} 25^{-2} + \log_5 5^4$
$= -2 + 4 = 2$.

35)

Area $= \frac{1}{2} \times \log_5 16 \times \log_5 64$
$= \frac{1}{2}\log_5 16 \times \log_5 16^{\frac{3}{2}} = \frac{1}{2}\log_5 16 \times \frac{3}{2}\log_5 16$
$= \frac{3}{4}(\log_5 16)^2 > \frac{3}{4} = 0.75$ as $\log_5 16 > 1$.

36)

$4^x - 7 \times 2^{x+1} + 45 = 0$
$\Rightarrow 2^{2x} - 7 \times 2^x \times 2^1 + 45 = 0$
$\Rightarrow (2^x)^2 - 14 \times 2^x + 45 = 0$
$\Rightarrow y^2 - 14y + 45 = 0$
$\Rightarrow (y - 9)(y - 5) = 0 \Rightarrow y = 9$ or $y = 5$
So $2^x = 9 \Rightarrow x = \log_2 9$ or $2^x = 5 \Rightarrow x = \log_2 5$.

37)

$\log_4 3 \times \log_4 9 \times \log_4 81$
$= \log_4 3 \times 2\log_4 3 \times 4\log_4 3 = 8(\log_4 3)^3 < 8$ as $\log_4 3 < 0$.

38)

$16^x + 2^{2x+1} - 99 = 0 \Rightarrow 2^{4x} + 2^{2x} \times 2^1 - 99 = 0$
$\Rightarrow (2^{2x})^2 + 2 \times 2^{2x} - 99 = 0 \Rightarrow y^2 + 2y - 99 = 0$
$\Rightarrow (y + 11)(y - 9) = 0 \Rightarrow y = -11$ or $y = 9$
$2^{2x} \neq -11$, So $2^{2x} = 9 \Rightarrow 2x = \log_2 9$
$\Rightarrow x = \frac{1}{2}\log_2 9 = \log_2 3$. $2^{2x} \neq -11$ as $2^{2x} > 0$.

39 a)

$y = 2^x : y^2 - 8y + 15 = 0$
$\Rightarrow (y - 5)(y - 3) = 0 \Rightarrow y = 5$ or $y = 3$
$2^x = 5 \Rightarrow x = \log_2 5$
$2^x = 3 \Rightarrow x = \log_2 3$.

39 b)

$y = 3^x : y^2 - 9y + 20 = 0$
$\Rightarrow (y - 4)(y - 5) = 0 \Rightarrow y = 4$ or $y = 5$
$3^x = 4 \Rightarrow x = \log_3 4$
$3^x = 5 \Rightarrow x = \log_3 5$.

39 c)

$y = 4^x : y^2 - 6y - 7 = 0$
$\Rightarrow (y - 7)(y + 1) = 0 \Rightarrow y = 7$ or $y = -1$
$4^x = 7 \Rightarrow x = \log_4 7$
$4^x \neq -1$.

39 d)

$y = 3^x : y^2 - 6y - 40 = 0$
$\Rightarrow (y - 10)(y + 4) = 0 \Rightarrow y = 10$ or $y = -4$
$3^x = 10 \Rightarrow x = \log_3 10$
$3^x \neq -4$.

40 a)

$2^{2x} - 5 \times 2^x \times 2^1 + 21 = 0$
$\Rightarrow (2^x)^2 - 10 \times 2^x + 21 = 0$
$y = 2^x : y^2 - 10y + 21 = 0$
$\Rightarrow (y - 3)(y - 7) = 0 \Rightarrow y = 3$ or $y = 7$
$2^x = 3 \Rightarrow x = \log_2 3$
$2^x = 7 \Rightarrow x = \log_2 7$.

40 b)

$5^{2x} - 3 \times 5^x \times 5^1 + 50 = 0$
$\Rightarrow (5^x)^2 - 15 \times 5^x + 50 = 0$
$y = 5^x : y^2 - 15y + 50 = 0$
$\Rightarrow (y - 5)(y - 10) = 0 \Rightarrow y = 5$ or $y = 10$

$5^x = 5 \Rightarrow x = 1$
$5^x = 10 \Rightarrow x = \log_5 10.$

40 c)

$2^{4x} + 2^{2x} \times 2^1 - 15 = 0$
$\Rightarrow (2^{2x})^2 + 2 \times 2^{2x} - 15 = 0$
$y = 2^{2x} : y^2 + 2y - 15 = 0$
$\Rightarrow (y+5)(y-3) = 0 \Rightarrow y = -5$ or $y = 3$
$2^{2x} \neq -5$
$2^{2x} = 3 \Rightarrow 2x = \log_2 3 \Rightarrow x = \frac{1}{2}\log_2 3.$

40 d)

$2^{4x} - 5 \times 2^{2x} \times 2^1 + 2^{2x} \times 2^3 - 80 = 0$
$\Rightarrow (2^{2x})^2 - 2 \times 2^{2x} - 80 = 0$
$y = 2^{2x} : y^2 - 2y - 80 = 0$
$\Rightarrow (y-10)(y+8) = 0 \Rightarrow y = 10$ or $y = -8$
$2^{2x} = 10 \Rightarrow 2x = \log_2 10 \Rightarrow x = \frac{1}{2}\log_2 10$
$2^{2x} \neq -8.$

41)

$8^x - 2^{2x+2} + 3 \times 2^x = 0$
$\Rightarrow 2^{3x} - 4 \times 2^{2x} + 3 \times 2^x = 0$
$\Rightarrow (2^x)^3 - 4 \times (2^x)^2 + 3 \times 2^x = 0$
$y = 2^x : y^3 - 4y^2 + 3y = 0 \Rightarrow y(y^2 - 4y + 3) = 0$
$\Rightarrow y(y-3)(y-1) = 0 \Rightarrow y = 0$ or $y = 3$ or $y = 1$
$2^x = 0$ has no real solutions
$2^x = 3 \Rightarrow x = \log_2 3$
$2^x = 1 \Rightarrow x = 0.$

42 a)

$3^{2x} \times 3^1 - 13 \times 3^x - 10 = 0$
$\Rightarrow 3(3^x)^2 - 13 \times 3^x - 10 = 0$
$y = 3^x : 3y^2 - 13y - 10 = 0$
$\Rightarrow (3y+2)(y-5) = 0 \Rightarrow y = -\frac{2}{3}$ or $y = 5$
$3^x \neq -\frac{2}{3}$
$3^x = 5 \Rightarrow x = \log_3 5.$

42 b)

$4^{2x} \times 16^1 - 139 \times 4^x - 45 = 0$
$\Rightarrow 16(4^x)^2 - 139 \times 4^x - 45 = 0$
$y = 4^x : 16y^2 - 139y - 45 = 0$
$\Rightarrow (16y+5)(y-9) = 0 \Rightarrow y = -\frac{5}{16}$ or $y = 9$
$4^x \neq -\frac{5}{16}$
$4^x = 9 \Rightarrow x = \log_4 9.$

42 c)

$2^{4x} \times 4^3 - 9 \times 2^{2x} \times 2^5 = 0$
$\Rightarrow 64(2^{2x})^2 - 288 \times 2^{2x} = 0$
$y = 2^{2x} : 64y^2 - 288y = 0$

$\Rightarrow 64y(y-\frac{9}{2})=0 \Rightarrow y=0$ or $y=\frac{9}{2}$
$2^{2x}\neq 0$
$2^{2x}=\frac{9}{2} \Rightarrow 2x=\log_2\frac{9}{2} \Rightarrow x=\frac{1}{2}\log_2\frac{9}{2}$.

42 d)

$2^{4x}\times 2^7+2^{2x}\times 2^3-3\times 2^{2x}\times 4^2-3=0$
$\Rightarrow 128(2^{2x})^2-40\times 2^{2x}-3=0$
$y=2^{2x}: 128y^2-40y-3=0$
$\Rightarrow (16y+1)(8y-3)=0 \Rightarrow y=-\frac{1}{16}$ or $y=\frac{3}{8}$
$2^{2x}\neq -\frac{1}{16}$
$2^{2x}=\frac{3}{8} \Rightarrow 2x=\log_2\frac{3}{8} \Rightarrow x=\frac{1}{2}\log_2\frac{3}{8}$.

43)

$$5^{x-3}=2^{x+4}$$
$$\log_5 5^{x-3}=\log_5 2^{x+4}$$
$$x-3=(x+4)\log_5 2$$
$$x-3=x\log_5 2+4\log_5 2$$
$$x-x\log_5 2=4\log_5 2+3$$
$$x(1-\log_5 2)=4\log_5 2+3$$
$$x=\frac{4\log_5 2+3}{1-\log_5 2}$$

44)

$$8^{x+2}=3^{2-x}$$
$$\log_3 8^{x+2}=\log_3 3^{2-x}$$
$$(x+2)\log_3 8=2-x$$
$$x\log_3 8+2\log_3 8=2-x$$
$$x\log_3 8+x=2-2\log_3 8$$
$$x(\log_3 8+1)=2-2\log_3 8$$
$$x=\frac{2-2\log_3 8}{\log_3 8+1}$$

45)

$$
\begin{gathered}
3^{x+1} = 8^{x-1} \\
\log_{10} 3^{x+1} = \log_{10} 8^{x-1} \\
(x+1)\log_{10} 3 = (x-1)\log_{10} 8 \\
x\log_{10} 3 + \log_{10} 3 = x\log_{10} 8 - \log_{10} 8 \\
\log_{10} 8 + \log_{10} 3 = x\log_{10} 8 - x\log_{10} 3 \\
\log_{10} 8 + \log_{10} 3 = x(\log_{10} 8 - \log_{10} 3) \\
\frac{\log_{10} 8 + \log_{10} 3}{\log_{10} 8 - \log_{10} 3} = x
\end{gathered}
$$

46 a)

$$
\begin{gathered}
3^{x} = 4^{x-1} \\
\log_{3} 3^{x} = \log_{3} 4^{x-1} \\
x = (x-1)\log_{3} 4 \\
x = x\log_{3} 4 - \log_{3} 4 \\
x - x\log_{3} 4 = -\log_{3} 4 \\
x(1 - \log_{3} 4) = -\log_{3} 4 \\
x = -\frac{\log_{3} 4}{1 - \log_{3} 4} = \frac{\log_{3} 4}{\log_{3} 4 - 1}
\end{gathered}
$$

46 b)

$$
\begin{gathered}
3^{x+5} = 5^{2-x} \\
\log_{3} 3^{x+5} = \log_{3} 5^{2-x} \\
x + 5 = (2-x)\log_{3} 5 \\
x + 5 = 2\log_{3} 5 - x\log_{3} 5 \\
x + x\log_{3} 5 = 2\log_{3} 5 - 5 \\
x(1 + \log_{3} 5) = 2\log_{3} 5 - 5 \\
x = \frac{2\log_{3} 5 - 5}{1 + \log_{3} 5}
\end{gathered}
$$

46 c)

$$8^{x-3} = 3^{x+8}$$
$$\log_3 8^{x-3} = \log_3 3^{x+8}$$
$$(x-3)\log_3 8 = x+8$$
$$x\log_3 8 - 3\log_3 8 = x+8$$
$$x\log_3 8 - x = 3\log_3 8 + 8$$
$$x(\log_3 8 - 1) = 3\log_3 8 + 8$$
$$x = \frac{3\log_3 8 + 8}{\log_3 8 - 1}$$

46 d)

$$10^{5-x} = 3^{x-6}$$
$$\log_3 10^{5-x} = \log_3 3^{x-6}$$
$$(5-x)\log_3 10 = x-6$$
$$5\log_3 10 - x\log_3 10 = x-6$$
$$5\log_3 10 + 6 = x\log_3 10 + x$$
$$5\log_3 10 + 6 = x(\log_3 10 + 1)$$
$$x = \frac{5\log_3 10 + 6}{\log_3 10 + 1}$$

47 a)

$$5^x = 9^{x+2}$$
$$\log_2 5^x = \log_2 9^{x+2}$$
$$x\log_2 5 = (x+2)\log_2 9$$
$$x\log_2 5 = x\log_2 9 + 2\log_2 9$$
$$x\log_2 5 - x\log_2 9 = 2\log_2 9$$
$$x(\log_2 5 - \log_2 9) = 2\log_2 9$$
$$x = \frac{2\log_2 9}{\log_2 5 - \log_2 9}$$

47 b)

$$
\begin{aligned}
3^{x-2} &= 7^{3-x} \\
\log_2 3^{x-2} &= \log_2 7^{3-x} \\
(x-2)\log_2 3 &= (3-x)\log_2 7 \\
x\log_2 3 - 2\log_2 3 &= 3\log_2 7 - x\log_2 7 \\
x\log_2 3 + x\log_2 7 &= 2\log_2 3 + 3\log_2 7 \\
x(\log_2 3 + \log_2 7) &= 2\log_2 3 + 3\log_2 7 \\
x &= \frac{2\log_2 3 + 3\log_2 7}{\log_2 3 + \log_2 7}
\end{aligned}
$$

47 c)

$$
\begin{aligned}
9^{x+4} &= 10^{4-x} \\
\log_2 9^{x+4} &= \log_2 10^{4-x} \\
(x+4)\log_2 9 &= (4-x)\log_2 10 \\
x\log_2 9 + 4\log_2 9 &= 4\log_2 10 - x\log_2 10 \\
x\log_2 9 + x\log_2 10 &= 4\log_2 10 - 4\log_2 9 \\
x(\log_2 9 + \log_2 10) &= 4\log_2 10 - 4\log_2 9 \\
x &= \frac{4\log_2 10 - 4\log_2 9}{\log_2 9 + \log_2 10}
\end{aligned}
$$

47 d)

$$
\begin{aligned}
5^{8-x} &= 6^{4-x} \\
\log_2 5^{8-x} &= \log_2 6^{4-x} \\
(8-x)\log_2 5 &= (4-x)\log_2 6 \\
8\log_2 5 - x\log_2 5 &= 4\log_2 6 - x\log_2 6 \\
x\log_2 6 - x\log_2 5 &= 4\log_2 6 - 8\log_2 5 \\
x(\log_2 6 - \log_2 5) &= 4\log_2 6 - 8\log_2 5 \\
x &= \frac{4\log_2 6 - 8\log_2 5}{\log_2 6 - \log_2 5}
\end{aligned}
$$

48 a)

$2\log_{10}(2-x) = \log_{10}(28-9x)$
$\Rightarrow \log_{10}(2-x)^2 = \log_{10}(28-9x)$
$\Rightarrow (2-x)^2 = 28-9x \Rightarrow 4-4x+x^2 = 28-9x$
$\Rightarrow x^2+5x-24=0.$

48 b)

$x^2+5x-24=0 \Rightarrow (x+8)(x-3)=0 \Rightarrow x=-8$ or $x=3$
$x \neq 3$ as $\log_{10}(2-3) = \log_{10}(-1)$ is undefined.
$x=-8$ is the only solution.

49)

$\log_5(8x-1) - \log_5(2-x) = 2 \Rightarrow \log_5(\frac{8x-1}{2-x}) = 2$
$\Rightarrow \frac{8x-1}{2-x} = 25 \Rightarrow 8x - 1 = 50 - 25x \Rightarrow 33x = 51$
$\Rightarrow x = \frac{51}{33}$.

50 a)

The first error is using the power rule incorrectly from line 1 to line 2. $3(\log_8 x)^2 \neq 6\log_8 x$
The second error is 'unlogging' from line 4 to line 5. If $\log_8 x = \frac{20}{17}$ then $x = 8^{\frac{20}{17}}$, not $x = (\frac{20}{17})^8$.

50 b)

$3(\log_8 x)^2 + 11\log_8 x - 20 = 0$
Let $y = \log_8 x : 3y^2 + 11y - 20 = 0$
$\Rightarrow (3y-4)(y+5) = 0 \Rightarrow y = \frac{4}{3}$ or $y = -5$
$\log_8 x = \frac{4}{3} \Rightarrow x = 8^{\frac{4}{3}}$
$\log_8 x = -5 \Rightarrow x = 8^{-5}$.

51 a)

The first error is dividing the 1 by the x from line 1 to line 2. $\frac{1}{\log_3 x} \neq \log_3 \frac{1}{x}$
The second error is from mistaking $\log_3 3 = 1$ for $\log_3 1 = 0$ line 4 to line 5. If George's working had been correct up to this point, the final answer would be $x = 1$ and not $x = 3$.

51 b)

$\log_3 \frac{1}{x} + \frac{1}{4\log_3 x} = 0 \Rightarrow \log_3 x^{-1} + \frac{1}{4\log_3 x} = 0$
$\Rightarrow -\log_3 x + \frac{1}{4\log_3 x} = 0 \Rightarrow \frac{1}{4\log_3 x} = \log_3 x$
$\Rightarrow \frac{1}{4} = (\log_3 x)^2 \Rightarrow \log_3 x = \pm\frac{1}{2} \Rightarrow x = 3^{\frac{1}{2}}$ or $x = 3^{-\frac{1}{2}}$
So $x = \sqrt{3}$ or $x = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$.

52 a)

$2\log_3 x = \log_3(mx-4) + 4$
$\Rightarrow \log_3 x^2 = \log_3(mx-4) + \log_3 3^4$
$\Rightarrow \log_3 x^2 = \log_3(81mx - 324)$
$\Rightarrow x^2 = 81mx - 324 \Rightarrow x^2 - 81mx + 324 = 0.$

52 b)

Only one root $\Rightarrow b^2 - 4ac = 0$
$\Rightarrow (-81m)^2 - 4 \times 1 \times 324 = 0$
$\Rightarrow 6561m^2 - 1296 = 0 \Rightarrow m^2 = \frac{16}{81} \Rightarrow m = \pm\frac{4}{9}$
As m is positive, $m = \frac{4}{9}$.

53)

$2^{2^x} = 5 \Rightarrow 2^x = \log_2 5 \Rightarrow x = \log_2(\log_2 5)$.

54)

$\log_5(6x-15)+1=0 \Rightarrow \log_5(6x-15)=-1$
$\Rightarrow 6x-15=5^{-1} \Rightarrow 6x=\frac{76}{5} \Rightarrow x=\frac{38}{15}$
$\therefore A(\frac{38}{15},0)$
$2\log_5(x-3)=0 \Rightarrow x-3=1 \Rightarrow x=4$
$\therefore B(4,0)$
$\log_5(6x-15)+1=2\log_5(x-3)$
$\Rightarrow \log_5(6x-15)+\log_5 5=\log_5(x-3)^2$
$\Rightarrow \log_5(30x-75)=\log_5(x-3)^2$
$\Rightarrow 30x-75=(x-3)^2 \Rightarrow 30x-75=x^2-6x+9$
$\Rightarrow x^2-36x+84=0 \Rightarrow x=18+4\sqrt{15}$
$\therefore C(18+4\sqrt{15}, 2\log_5(15+4\sqrt{15}))$.

55)

$2\log_4(x+6)=\log_4(x+48)-\log_4(2-x)$
$\Rightarrow \log_4(x+6)^2=\log_4(\frac{x+48}{2-x}) \Rightarrow (x+6)^2=\frac{x+48}{2-x}$
$\Rightarrow x^2+12x+36=\frac{x+48}{2-x}$
$\Rightarrow 2x^2+24x+72-x^3-12x^2-36x=x+48$
$\Rightarrow 0=x^3+10x^2+13x-24$
$\Rightarrow (x+8)(x+3)(x-1)=0$
$\Rightarrow x=-8$ or $x=-3$ or $x=1$
$x\neq -8$ as $\log_4(-8+6)=\log_4(-2)$ which is not real.
When $x=-3$, $y=2\log_4(-3+6)=\log_4 9$
When $x=1$, $y=2\log_4(1+6)=\log_4 49$
So $A(1,\log_4 49)$ and $B(-3,\log_4 9)$
$2\log_4(x+6)=0 \Rightarrow x=-5$, so $C(-5,0)$.